NGUYỄN MINH PHƯƠNG

AI Engineer

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TECHNICAL SKILLS

Programming: Python, Java, SQL, TypeScript, JavaScript
AI / Computer Vision: PyTorch, Hugging Face Transformers, PEFT, QLoRA, Vision-Language Models, Multi-label Image Classification, Grad-CAM, scikit-learn, NumPy, pandas, Matplotlib
LLM: RAG, Embeddings, Prompt Engineering, MedGemma, Ollama, Milvus
Model Evaluation: AUROC, AUPRC, F1, Precision / Recall, Threshold Calibration, Bootstrap CIs, Ablation Studies, BLEU, ROUGE-L, BERTScore
Backend & Deployment: FastAPI, Spring Boot 3, REST API, PostgreSQL, Docker, Docker Compose, GCP, Linux, CUDA
Tools: Git, GitHub, pytest, W&B, Jira, AI Agents (Claude Code, Codex, GitHub Copilot)

KEY PROJECTS

META-CXR - Multimodal AI for Chest X-ray Report Generation | Graduation thesis • Solo GitHub

- Reproduced and extended a two-stage vision-language model (IEEE Access 2025) in PyTorch on MIMIC-CXR: two medical vision encoders (BioViL-T, PubMedCLIP) + Q-Former classify 14 abnormalities, then MedGemma-4B fine-tuned with QLoRA writes the report.
- Model optimization: partially unfreezing both encoders raised test macro-AUROC 0.744 $\rightarrow$ 0.764 on 3,269 held-out studies (paired bootstrap CI above 0, gains on 14/14 labels), with thresholds calibrated on validation only.
- Fixed an image-representation bug: restored each encoder's native patch grid and removed a constant component by per-image centering, cutting pairwise patch cosine similarity from 0.674 to -0.014 at only +0.6% step time.
- Reproducible experiments: controlled A/B runs dropped components with no measurable gain (Grad-CAM explanation loss, deeper unfreezing); joining MS-CXR boxes by image ID exposed 166 boxes leaking into the test split; 1,012 pytest tests passed.
- LLM debugging: found the chat stop token was overridden, so 78% of generated tokens came after the model had finished.
- Infrastructure: full training on one 16 GB GPU (fixed recurring kernel faults traced to unstable RAM settings); multi-GPU DDP smoke runs on 2$\times$ T4 with checkpoints on Google Cloud Storage.

FPT Tri Thuc - Enterprise RAG Assistant | Personal project • Solo GitHub

- Python ingestion pipeline (PDF / web / text loaders, chunking, embedding, hash dedup) into Milvus 2.4, plus a Spring Boot RAG API for Vietnamese Q&A over FPT corporate documents.
- Pulls 64 candidates from Milvus, re-ranks by cosine plus keyword / bigram coverage, cites sources and abstains when evidence is weak; shows each chunk's cosine score in the UI so retrieval quality can be audited.
- Token streaming over SSE with JSON fallback and per-upstream timeouts; 25 JUnit / Mockito tests and 9 Playwright E2E tests; runs on Docker Compose.

Chinese Learning Platform - Full-Stack AI Web App | Personal project • Solo GitHub

- Designed, built and deployed the whole product alone in under 2 months; used by friends, and iterated on their feedback.
- Integrated AI into a real product: a Spring Boot API orchestrates an LLM (chat, translation, writing feedback), Azure Speech and a Python FastAPI service for F0 pitch analysis (per-syllable tone scoring); PostgreSQL + Flyway, Docker, deployed on GCP over HTTPS.

WORK EXPERIENCE

FPT Software | Java Developer Intern Ho Chi Minh City | May 2026 - Oct 2026

Project BUHINH - modernizing a Japanese automaker's material management (BOM) & production-cost estimation system

- Migrated legacy Java EE sub-systems to Spring Boot / Spring MVC as Maven multi-module projects, converting each layer with GitHub Copilot under the team's prompt rules and hand-reviewing every pattern the AI flagged.
- Verified migrated JSP / JavaScript screens for functional parity and traced data errors through SQL trace logs; Scrum sprints on Jira, GitLab merge requests reviewed by the team lead.

EDUCATION

University of Information Technology (UIT) - VNU-HCM | Information Systems 2023 - 2027

GPA: 8.5/10 | Expected graduation: Jun 2027

AWARDS AND CERTIFICATES

- Outstanding Student - UIT - VNU-HCM, Semester 2 of academic year 2024 - 2025 2025
- Academic Scholarship - UIT - VNU-HCM, Top 8% of major, awarded for 2 semesters 2023 - 2025

LANGUAGES

English: B2 (CEFR) equivalent - reads technical documentation and AI/ML research papers with ease